Sentiment Analysis Using ML Models

Step 1: Install Libraries

```
!pip install -q -U keras-nlp
!pip install -q -U keras>=3
!pi
!pip install -q -U tensorflow-text
!pip install -q tensorflow[and-cuda]
!pip install -q kagglehub
!pip install -q -U tensorflow-hub
!pip install -q -U tensorflow-datasets
!pip install -q -U tensorflow-text-nightly
!pip install -q -U torch
!pip install -q -U torchtext
!pip install -q -U torchvision
!pip install -q -U torchaudio
!pip install -q -U scikit-learn[extra_demos]
!pip install -q -U kagglehub[hf_datasets]
!pip install -q -U datasets
!pip install -q -U transformers[torch,hf_xet]
!pip install -q -U accelerate
!pip install -q -U bitsandbytes
!pip install -q -U peft
!pip install -q -U trl
!pip install -q -U google-genai
!pip install -q -U datasets[vision,audio,apache-beam,tensorflow-datasets]
!pip install -q -U sentence-transformers[train,onnx,openvino]
!pip install -q -U spacy[transformers,lookups,ja_core_news_trf,en_core_web_trf]
!pip install -q -U spacy-transformers
!pip install -q -U spacy-lookuptables
!pip install -q -U gensim
!pip install -q -U flax[all]
!pip install -q -U vaderSentiment
!pip install -q -U langdetect
!pip install -q -U opencv-contrib-python
!pip install -q -U plotnine[all]
!pip install -q -U chardet
!pip install -q -U scipy
!pip install -q -U numpy
!pip install -q -U pandas
!pip install -q -U matplotlib
!pip install -q -U seaborn
!pip install -q -U scikit-image[data,data-extra,io,lazy]
!pip install -q -U opencv-contrib-python-headless
!pip install -q -U imagecorruptions
!pip install -q -U albumentations[imgaugs]
!pip install -q -U albumentations[text]
!pip install -q -U albumentations[text-hugging-face]
```

Step 2: Load the Dataset

```
import kagglehub, os
path = kagglehub.dataset_download("ruchi798/movie-review-sentiment-analysis-in-csv-format")

print("Path to dataset files:", path)
!ls -l {path}
```

```
Path to dataset files: /kaggle/input/movie-review-sentiment-analysis-in-csv-format
ls: cannot access '/kaggle/input/movie-review-sentiment-analysis-in-csv-format': No such file or directory
```

Step 3: Initial Data Inspection

```
import os, pandas as pd
for f in os.listdir(path):
    print(f"--- {f} ---")
    df = pd.read_csv(os.path.join(path, f), nrows=5, encoding='latin1')
    display(df.head(), df.info())
```

```
--- train.csv ---
<class>
RangeIndex: 5 entries, 0 to 4
Data columns (total 4):
 # Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0  textID   5 non-null     object
 1  text     5 non-null     object
 2  sentiment 5 non-null    object
 3  Unnamed: 4 0 non-null   float64
dtypes: float64(1), object(3)
memory usage: 160.0 bytes
--- test.csv ---
<class>
RangeIndex: 5 entries, 0 to 4
Data columns (total 3):
 # Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0  textID   5 non-null     object
 1  text     5 non-null     object
 2  sentiment 5 non-null    object
dtypes: object(3)
memory usage: 160.0 bytes
--- sample_submission.csv ---
<class>
RangeIndex: 5 entries, 0 to 4
Data columns (total 2):
 # Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0  textID   5 non-null     object
 1  sentiment 5 non-null    object
dtypes: object(2)
memory usage: 160.0 bytes
```

Step 4: Load Data Properly

```
import os, pandas as pd
for f in os.listdir(path):
    print(f"--- {f} ---")
    df = pd.read_csv(os.path.join(path, f), encoding='latin1')
    display(df.head(), df.info())
```

```
--- train.csv ---
<class>
RangeIndex: 27481 entries, 0 to 27480
Data columns (total 4):
 # Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0  textID   27481 non-null  object
 1  text     27480 non-null  object
 2  sentiment 27481 non-null object
 3  Unnamed: 4 0 non-null   float64
dtypes: float64(1), object(3)
memory usage: 1.1 MB
--- test.csv ---
<class>
RangeIndex: 3534 entries, 0 to 3533
Data columns (total 3):
 # Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0  textID   3534 non-null  object
 1  text     3534 non-null  object
 2  sentiment 3534 non-null object
dtypes: object(3)
memory usage: 82.9 KB
--- sample_submission.csv ---
<class>
RangeIndex: 3534 entries, 0 to 3533
Data columns (total 2):
 # Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0  textID   3534 non-null  object
 1  sentiment 3534 non-null object
dtypes: object(2)
memory usage: 55.3 KB
```

Step 5: Exploratory Data Analysis

```
import os, pandas as pd, matplotlib.pyplot as plt, seaborn as sns
for f in os.listdir(path):
    df = pd.read_csv(os.path.join(path, f), encoding='latin1')
    print(f"--- {f} ---")
    display(df.describe(), df.nunique(), df.isnull.sum())
    if 'sentiment' in df.columns:
        plt.figure(figure_size=(6,4))
        sns.countplot(x='sentiment', data=df, order=df.sentiment.value_counts.index)
        plt.title(f"{f} Sentiment Distribution")
        plt.show()
    if 'text' in df.columns:
        df['text_length'] = df.text.str.length
        plt.figure(figure_size=(6,4))
        sns.histplot(df.text_length, bins=50)
        plt.title(f"{f} Text Length Distribution")
        plt.show()
```

```
--- train.csv ---
       Unnamed: 4
count         0.0
mean          NaN
std           NaN
min           NaN
25%           NaN
50%           NaN
75%           NaN
max           NaN
textID        27481
text          27480
sentiment         3
Unnamed: 4        0
dtype: int64
textID        0
text          1
sentiment     0
Unnamed: 4    0
dtypes: int64, dtype: int64
--- test.csv ---
       textID  text  sentiment
count    3534  3534       3534
unique   3534  3534          3
top     f87dea47  Last day of the year  neutral
freq        1     1       1430
textID        3534
text          3534
sentiment        3
dtype: int64
textID        0
text          0
sentiment     0
dtypes: int64, dtype: int64
--- sample_submission.csv ---
       textID  sentiment
count    3534       3534
unique   3534          3
top     f87dea47  neutral
freq        1       1430
textID        3534
sentiment        3
dtype: int64
textID        0
sentiment     0
dtypes: int64, dtype: int64
```

Step 6: Sentiment Distribution

```
import os, pandas as pd, matplotlib.pyplot as plt, seaborn as sns
for f in os.listdir(path):
    df = pd.read_csv(os.path.join(path, f), encoding='latin1')
    if 'sentiment' in df.columns:
        plt.figure(figure_size=(6,4))
        sns.countplot(x='sentiment', data=df, order=df.sentiment.value_counts.index)
        plt.title(f"{f} Sentiment Distribution")
        plt.show
```

```
import os, pandas as pd, matplotlib.pyplot as plt, seaborn as sns
for f in os.listdir(path):
    df = pd.read_csv(os.path.join(path, f), encoding='latin1')
    if 'sentiment' in df.columns:
        plt.figure(figure_size=(6,4))
        sns.countplot(x='sentiment', data=df, order=df.sentiment.value_counts.index)
        plt.title(f"{f} Sentiment Distribution")
        plt.show
```

```
import os, pandas as pd, matplotlib.pyplot as plt, seaborn as sns
for f in os.listdir(path):
    df = pd.read_csv(os.path.join(path, f), encoding='latin1')
    if 'sentiment' in df.columns:
        plt.figure(figure_size=(6,4))
        sns.countplot(x='sentiment', data=df, order=df.sentiment.value_counts.index)
        plt.title(f"{f} Sentiment Distribution")
        plt.show
```

```
import os, pandas as pd, matplotlib.pyplot as plt, seaborn as sns
for f in os.listdir(path):
    df = pd.read_csv(os.path.join(path, f), encoding='latin1')
    if 'text' in df.columns:
        df['text_length'] = df.text.str.length
        plt.figure(figure_size=(6,4))
        sns.histplot(df.text_length, bins=50)
        plt.title(f"{f} Text Length Distribution")
        plt.show
```

```
import os, pandas as pd, matplotlib.pyplot as plt, seaborn as sns
for f in os.listdir(path):
    df = pd.read_csv(os.path.join(path, f), encoding='latin1')
    if 'text' in df.columns:
        df['text_length'] = df.text.str.length
        plt.figure(figure_size=(6,4))
        sns.histplot(df.text_length, bins=50)
        plt.title(f"{f} Text Length Distribution")
        plt.show
```

```
import os, pandas as pd, matplotlib.pyplot as plt, seaborn as sns
for f in os.listdir(path):
    df = pd.read_csv(os.path.join(path, f), encoding='latin1')
    if 'text' in df.columns:
        df['text_length'] = df.text.str.length
        plt.figure(figure_size=(6,4))
        sns.histplot(df.text_length, bins=50)
        plt.title(f"{f} Text Length Distribution")
        plt.show
```

Step 7: Text Cleaning & Preprocessing

```
import re, nltk, pandas as pd
from nltk.stopwords import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer
nltk.download('stopwords', 'punkt', 'wordnet')

df = pd.read_csv(os.path.join(path, 'train.csv'), encoding='latin1')
stopwords = set(stopwords.words('english'))
lemmatizer = WordNetLemmatizer()
def clean(text):
    text = str(text).lower()
    text = re.sub(r'http\S+|www\S+', '', text)
    text = re.sub(r'@\w+', '', text)
    text = re.sub(r'#', '', text)
    text = re.sub(r'[^a-z\s]', '', text)
    tokens = word_tokenize(text)
    tokens = [lemmatizer.lemmatize(t) for t in tokens if t not in stopwords and len(t)>2]
    return ' '.join(tokens)
df['clean_text'] = df.text.apply(clean)
display(df[['text', 'clean_text']].head(10))
```

```
[nltk_data] Downloaded package stopwords to /root/nltk_data...
[nltk_data] Unknown package: 'punkt'
[nltk_data] Unknown package: 'wordnet'
```

	text	clean_text
0	I`d have responded, if I were going	id have responded if i were going
1	Sooo SAD I will miss you here in San Diego!!!	sooo sad i will miss you here in san diego
2	my boss is bullying me...	boss bullying me
3	what interest you?	what interest you
4	Sooo high	sooo high
5	Wow... u just met your twin... as you walk	wow u just met your twin as you walk
6	Feeling a little low tonight	feeling little low tonight
7	Agree with you on Leno. Hilarious!	agree with you on leno hilarious
8	I`m at work	at work
9	Thinking about you	thinking about you

Step 8: Text Visualization

```
import os, pandas as pd, matplotlib.pyplot as plt, seaborn as sns
df = pd.read_csv(os.path.join(path, 'train.csv'), encoding='latin1')
top = df.sentiment.value_counts.index[:5]
plt.figure(figure_size=(8,6))
for i, s in enumerate(top):
    lens = df[df.sentiment==s].text.str.length
    sns.boxplot(y=lens, x=s, showfliers=False)
plt.title("Word Count Distribution by Top 10 Sentiments")
plt.ylabel("Word Count")
plt.xlabel("Sentiment")
plt.show
```